Fourier Analysis

Periodic Functions

Sketch, Even and Odd functions

Conducted By

Partho Sutra Dhor

Faculty, Mathematics and Natural Sciences

BRAC University, Dhaka, Bangladesh

Periodic Functions

Periodic Functions

A function $f(x)$ is said to be periodic if there is a $P > 0$ such that

$$f(x+P) = f(x), \qquad \text{for all } x$$

The least value of $P > 0$ is called the period.

$$\sin(x + \underline{2\pi}) = \sin(x)$$

$$\sin(x + \underline{4\pi}) = \sin(x)$$

$$\sin(x + \underline{6\pi}) = \sin x$$

Period $= \underline{\underline{2\pi}}$

Period of basic Trigonometric functions

$\sin(x) \longrightarrow$ Period $= 2\pi$

$\cos(x) \longrightarrow$ Period $= 2\pi$

$\tan(x) \longrightarrow$ Period $= \pi$

	Period
$\csc x$	2π
$\sec x$	2π
$\cot x$	π

Period of any Trigonometric functions

$a \cdot \sin^b(cx+d)+f \longrightarrow$ Period

$a \longrightarrow$ amplitude

$d \longrightarrow$ phase shift

$f \longrightarrow$

$c \longrightarrow$ New period $= \dfrac{\text{original}}{c}$

$b \longrightarrow$ even $\quad \underbrace{\sin \text{ or } \cos}_{\text{period } \pi}$

$b \longrightarrow$ odd $\quad \underbrace{\sin \text{ or } \cos}_{\text{period } 2\pi}$

PROBLEM Determine the period of the following function

$$f(x) = 1952\cos^{1971}(2025x + 2026)$$

$\sin x \longrightarrow$ Period $= 2\pi$

$\sin^{1971}(x) \longrightarrow$ Period $= 2\pi$

$\sin^{1971}(2025x) \longrightarrow$ Period $= \dfrac{2\pi}{2025}$ ✓

PROBLEM Determine the period of the following function

$$f(x) = \sin^{2024}(2025x + 2026)$$

$\sin(x) \longrightarrow$ Period 2π

$\sin^{2024}(x) \longrightarrow$ period $= \pi$

$\sin^{2024}(2025x) \longrightarrow$ period $= \dfrac{\pi}{2025}$

$\therefore \sin^{2024}(2025x + 2026) \longrightarrow$ Period $\dfrac{\pi}{2025}$.

PROBLEM Determine the period of the following function

$$f(x) = \tan^{2025}(2024x + 2023)$$

$\tan(x) \longrightarrow$ period $= \pi$

$\tan^{2025}(x) \longrightarrow$ period $= \pi$

$\tan^{2025}(2024x) \Longrightarrow$ period $= \dfrac{\pi}{2024}$

$\tan^{2025}(2024x + 2023) \longrightarrow$ period $= \dfrac{\pi}{2024}$

PROBLEM Determine the period of the following function

$$f(x) = \tan^{2024}(2024x + 2023)$$

$\frac{\pi}{2024}$.

$\tan(x) \longrightarrow$ period π

$\tan^{2024}(x) \longrightarrow$ period π

$\tan^{2024}(2024x) \longrightarrow$ period $\frac{\pi}{2024}$

Period of the sum of two Trigonometric functions

$$f(x) = \underbrace{\sin(6x)}_{\frac{2\pi}{6}} - \underbrace{\sin(8x)}_{\frac{2\pi}{8}}$$

$$\text{L.C.M}\left(\frac{\pi}{3}, \frac{\pi}{4}\right)$$

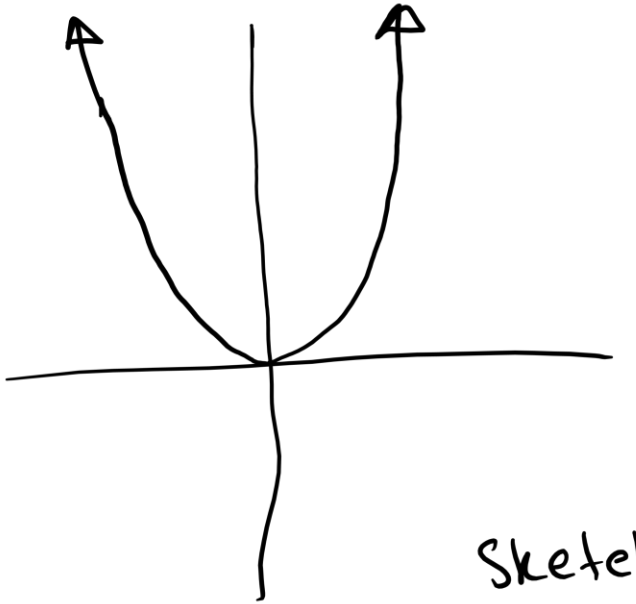
$$= \frac{\pi}{1}$$

$$= \pi$$

Sketch of Functions

Sketch vs Plot

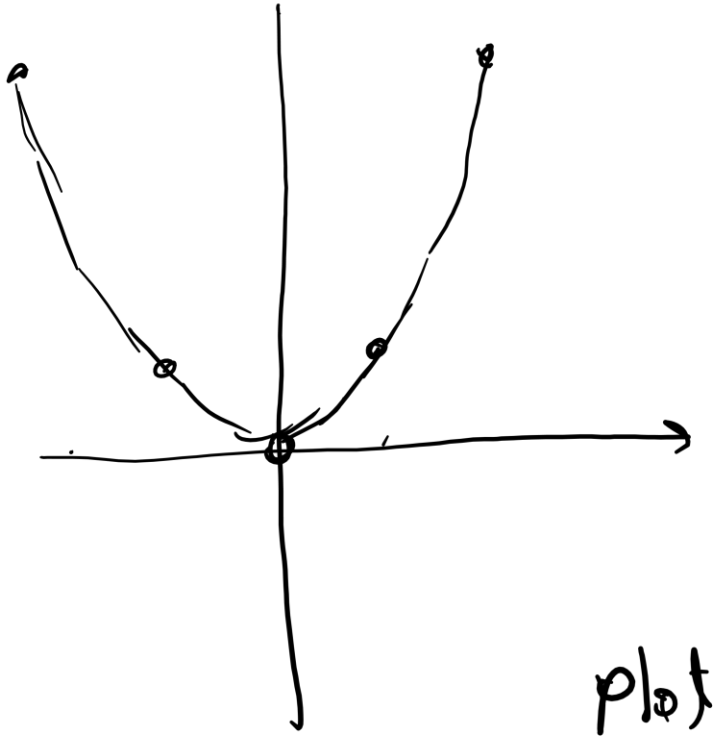
$y = x^2$



Sketch

$y = x^2$

x	y
0	0
1	1
2	4
-1	1
-2	4

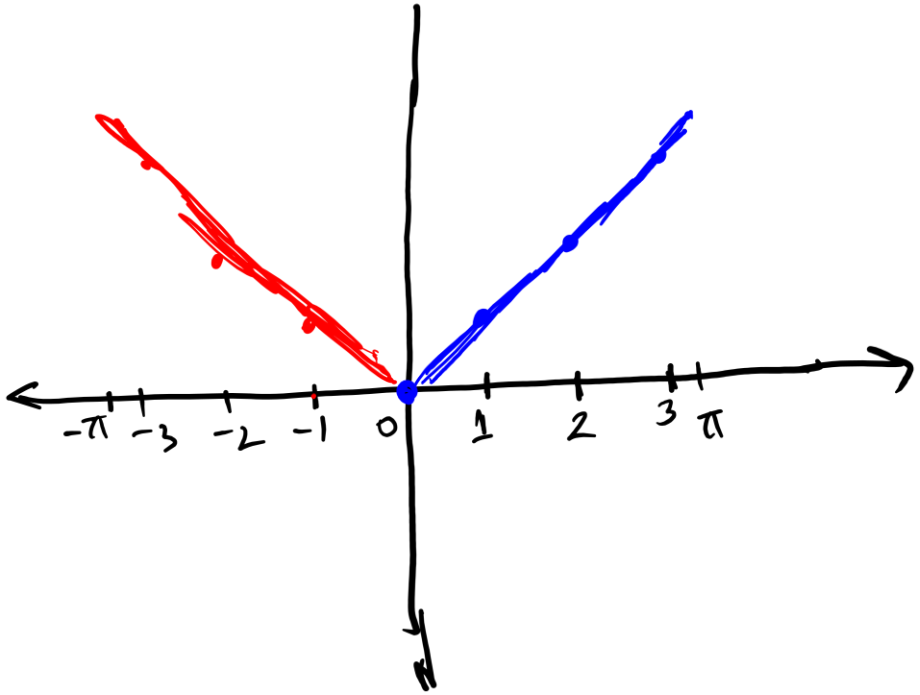


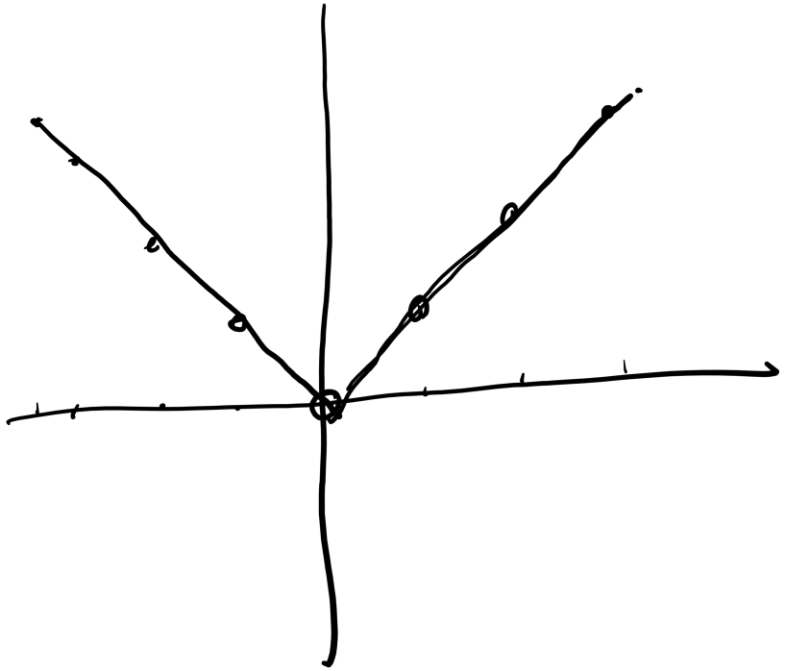
Plot

PROBLEM Sketch the function

V.V.g

$$f(x) = \begin{cases} -x & -\pi < x < 0 \\ x & 0 \le x < \pi \end{cases} \qquad or \qquad f(x) = |x|, \quad -\pi < x < \pi$$

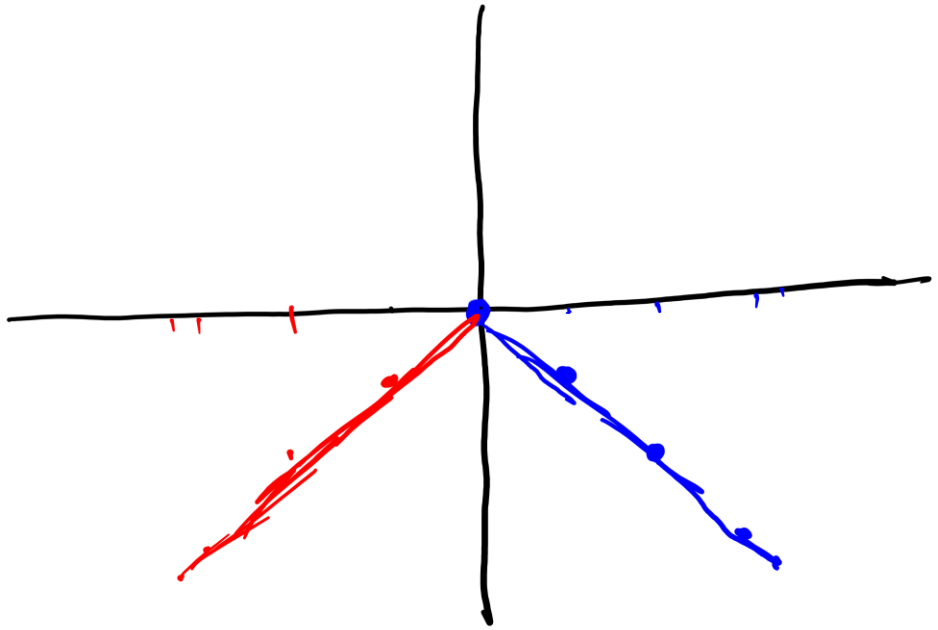


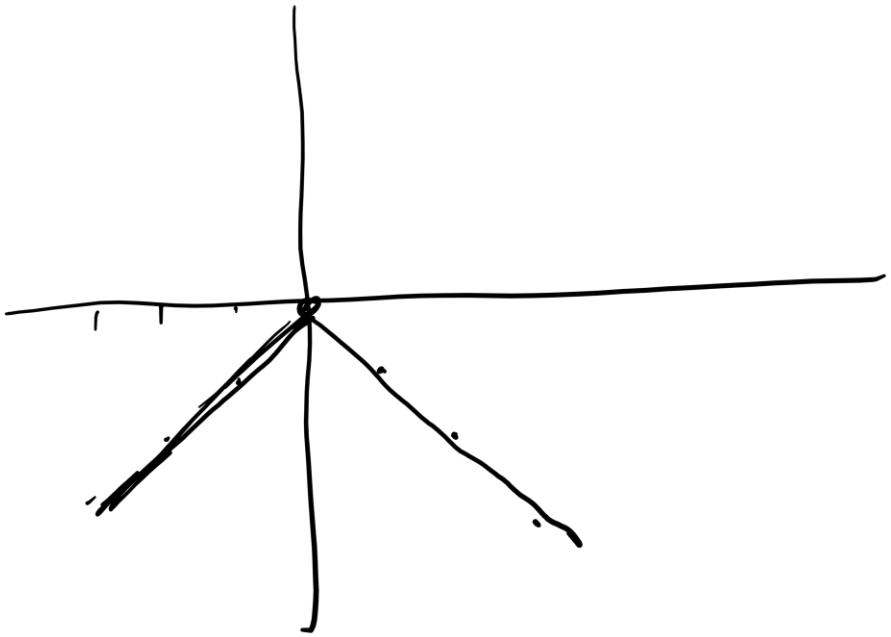


PROBLEM Sketch the function

v.v.g

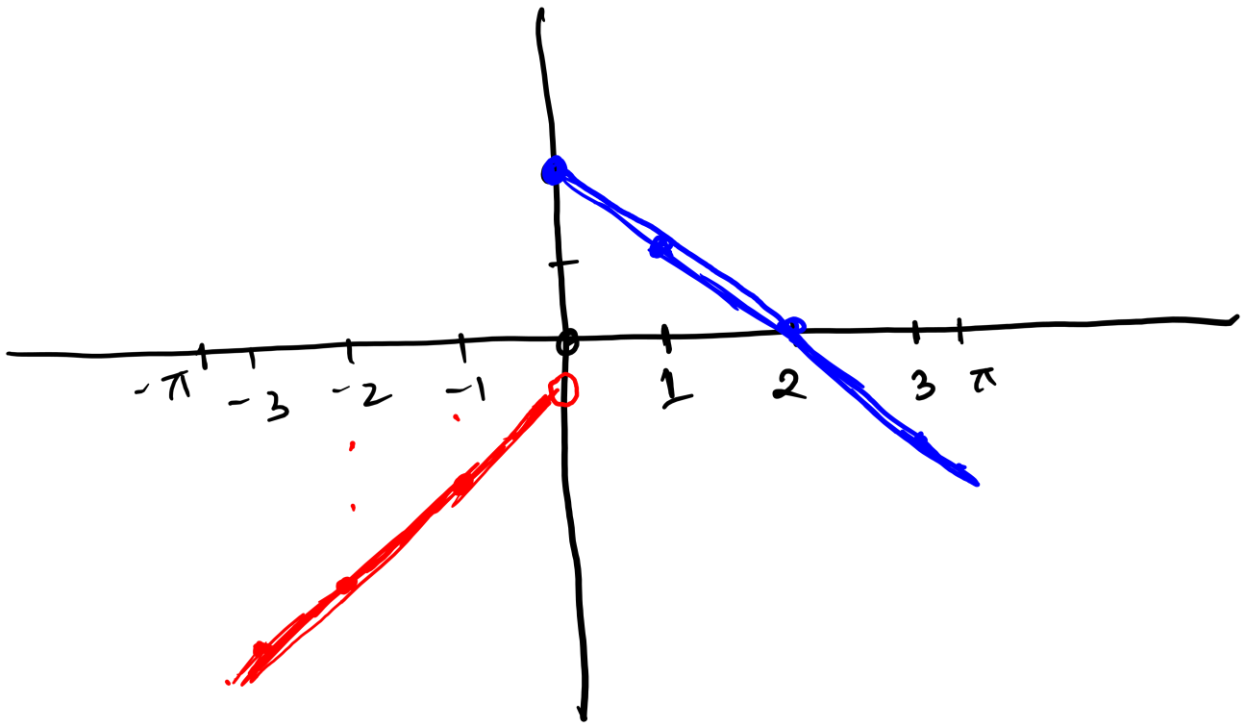
$$f(x) = \begin{cases} x & -\pi < x < 0 \\ -x & 0 \le x < \pi \end{cases} \qquad \text{or} \qquad f(x) = -|x|, \quad -\pi < x < \pi$$





PROBLEM Sketch the function

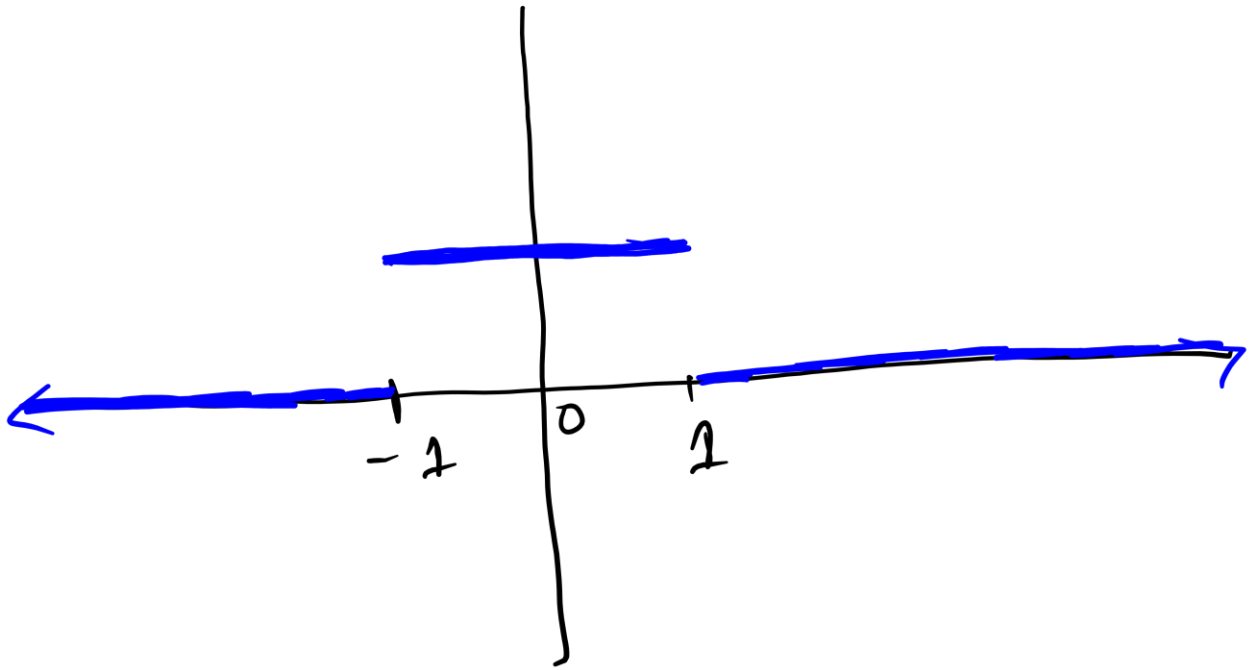
$$f(x) = \begin{cases} x - 1 & -\pi < x < 0 \\ -x + 2 & 0 \leq x < \pi \end{cases}$$



PROBLEM Sketch the function

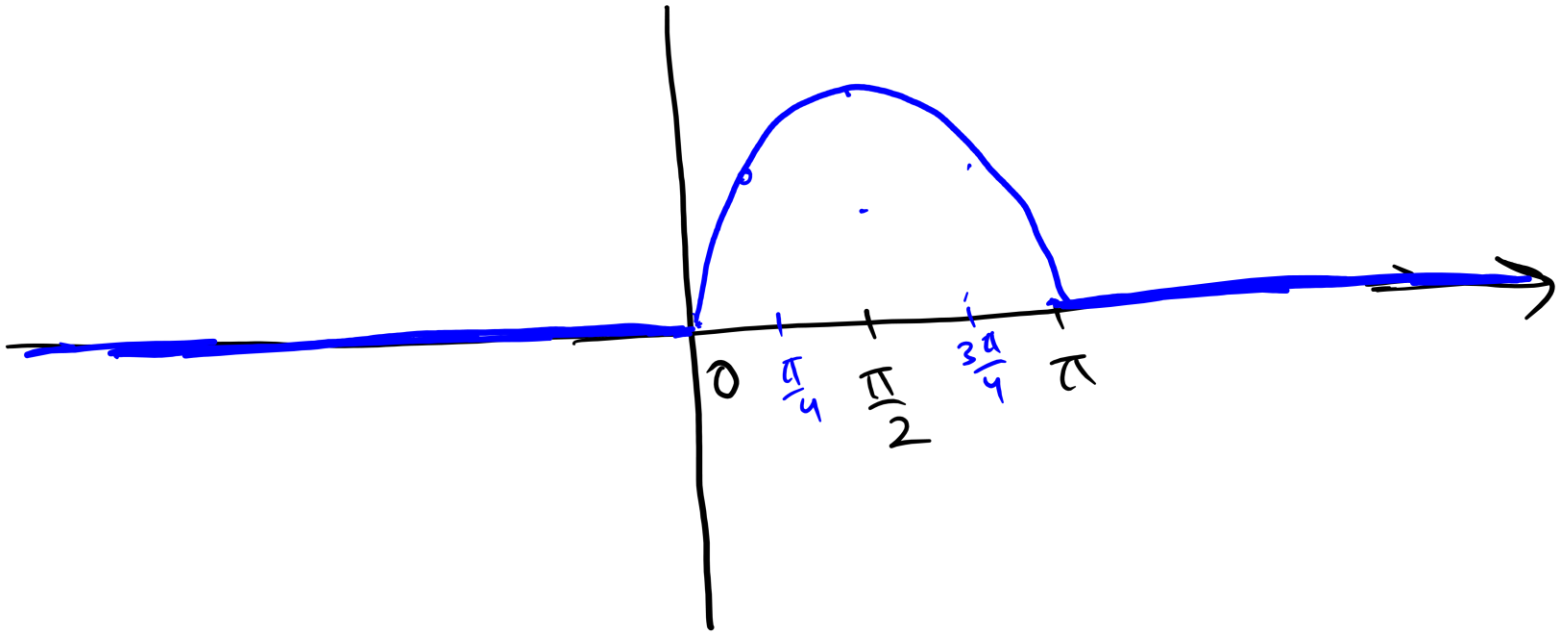
$$f(x) = \begin{cases} 1 & |x| < 1 \\ 0 & otherwise \end{cases}$$

$-1 < x < 1$



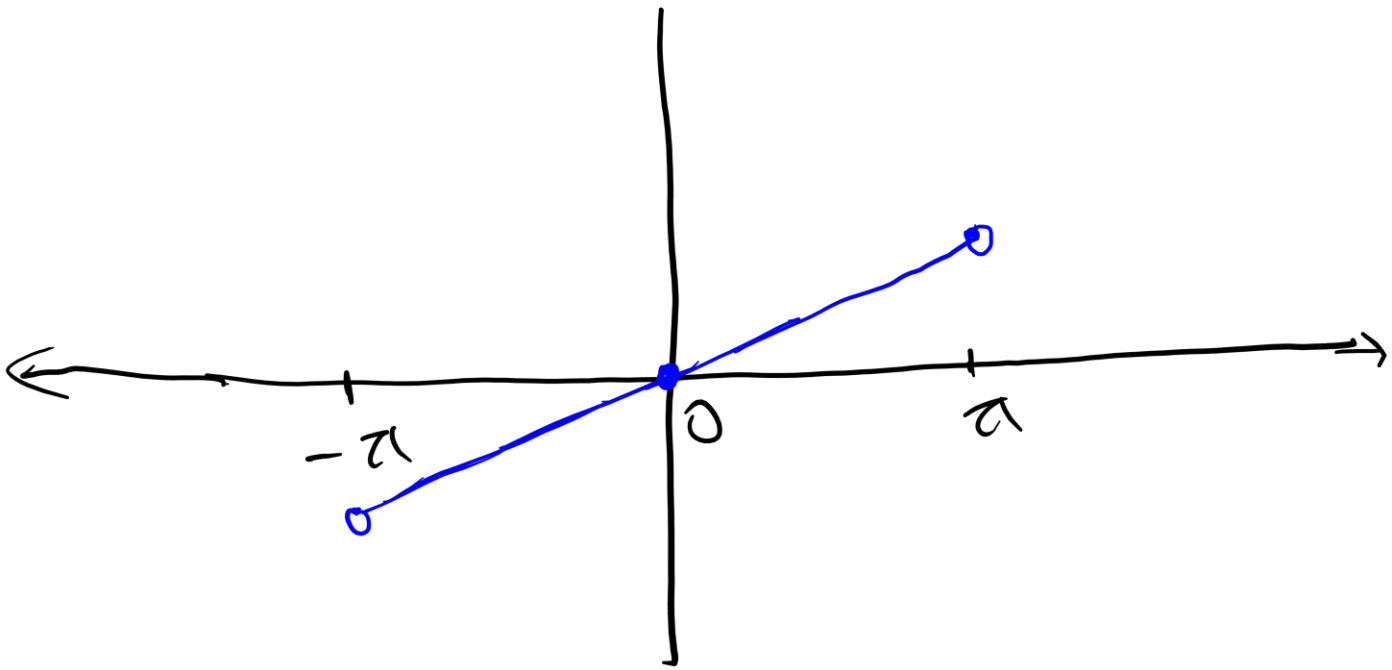
PROBLEM Sketch the function

$$f(x) = \begin{cases} 2\sin x & 0 < x < \pi \\ 0 & otherwise \end{cases}$$



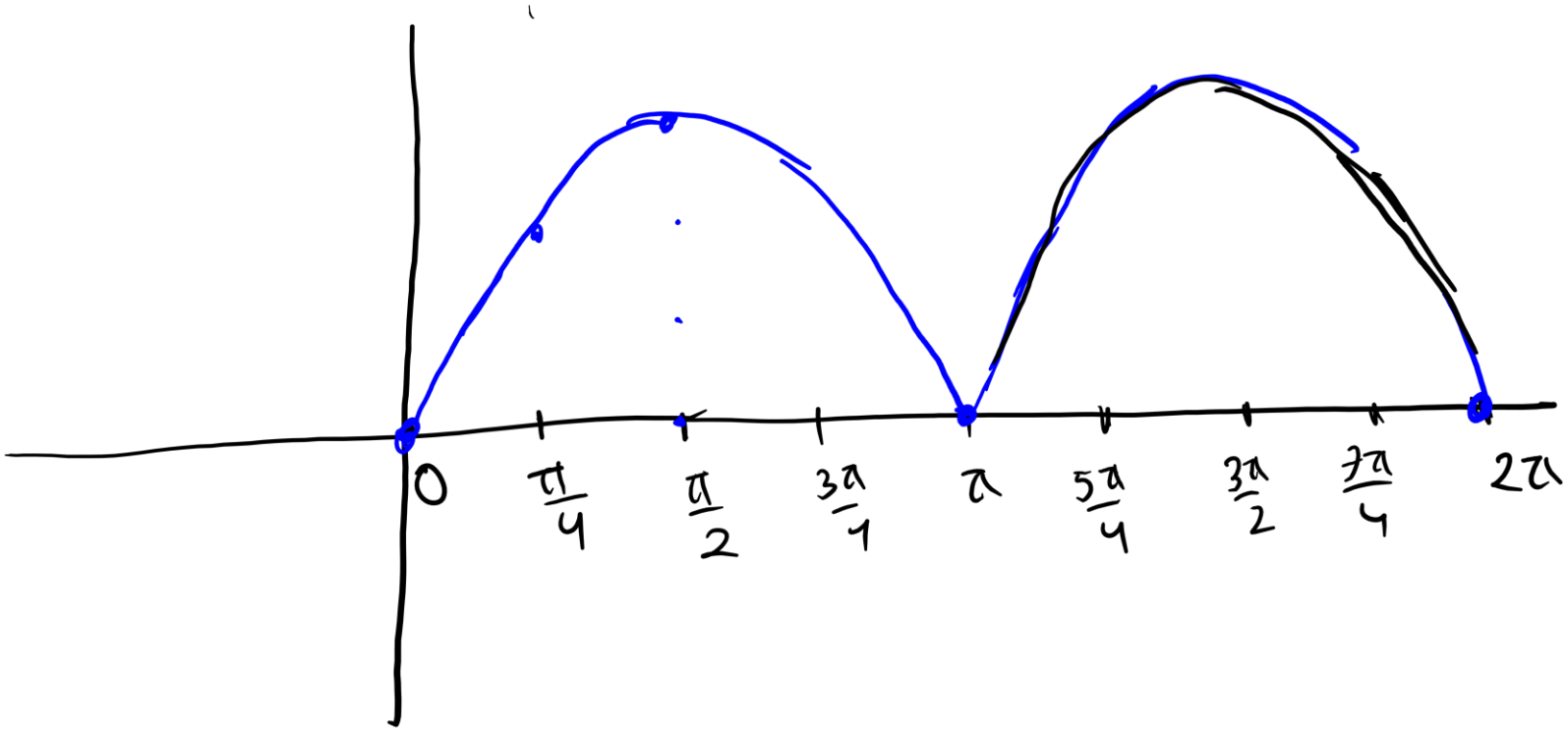
PROBLEM Sketch the function

$$f(x) = \frac{x}{\pi} \quad -\pi < x < \pi$$



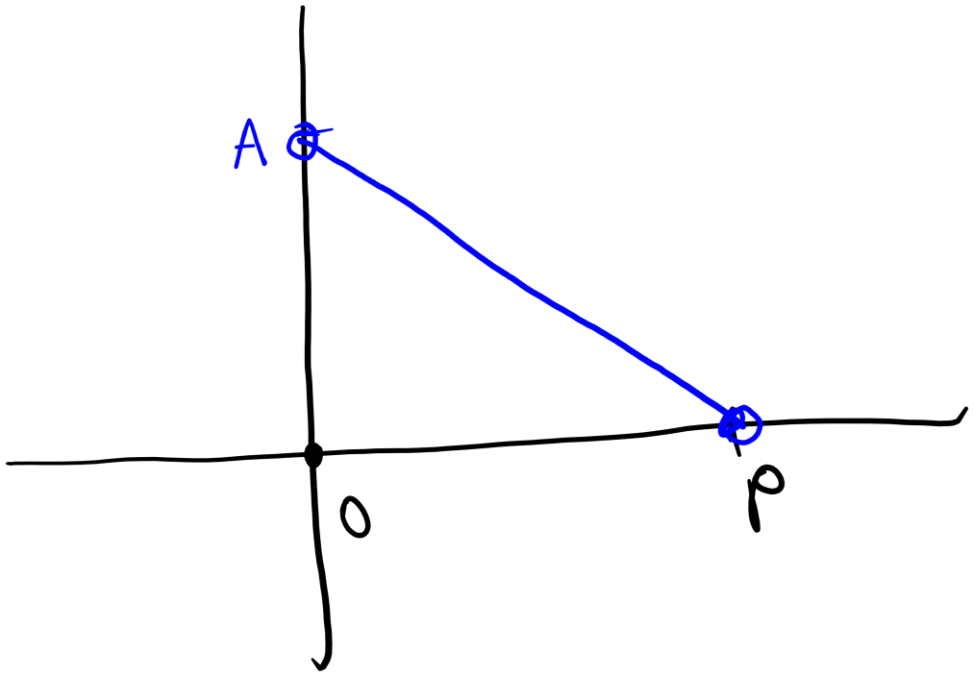
PROBLEM Sketch the function

$$f(x) = 3|\sin x| \quad 0 < x < 2\pi$$



PROBLEM Sketch the function

$$f(x) = A - \frac{Ax}{P} \quad 0 < x < P$$



Even and Odd functions

Definition of Even functions

function must be defined on $(-L, L)$ or $[-L, L]$

and $f(-x) = f(x)$

* Symmetric about y-axis.

Definition of Odd functions

function must be defined on $(-L, L)$ or $[-L, L]$

$$f(-x) = -f(x)$$

* $f(x)$ is symmetric about origin

A function can be neither even nor odd

PROBLEM Some Even Functions

1. $f(x) = x^4$ ✓

$f(-x) = (-x)^4 = x^4 = f(x)$

2. $f(x) = \cos(x)$ ✓

$f(-x) = \cos(-x) = \cos(x) = f(x)$

3. $f(x) = |x|$ ✓

$f(-x) = |-x| = |x| = f(x)$

4. $f(x) = 3$ ✓

$f(-x) = 3 = f(x)$

5. $f(x) = 3|\sin(x)|$ ✓

$f(-x) = 3 \cdot |\sin(-x)| = 3 \cdot |-\sin(x)| = 3 \cdot |\sin(x)|$
$= f(x)$

6. $f(x) = x^3 \sin(x)$ ✓

$f(-x) = (-x)^3 \cdot \sin(-x) = -x^3 \cdot (-) \sin(x)$
$= x^3 \sin(x) = f(x)$

PROBLEM Some Odd functions

1. $f(x) = x^3$ ✓

$f(-x) = (-x)^3 = -x^3 = -f(x)$

2. $f(x) = \sin(x)$ ✓

$f(-x) = \sin(-x) = -\sin(x) = -f(x)$

3. $f(x) = \frac{|x|}{x}$ ✓

$f(-x) = \frac{|-x|}{-x} = \frac{|x|}{-x} = -\frac{|x|}{x} = -f(x)$

4. $f(x) = \frac{x}{\pi}$ ✓

$f(x) = \frac{-x}{\pi} = -\frac{x}{\pi} = -f(x)$

5. $f(x) = x^3 \cos(x)$ ✓

$f(-x) = (-x)^3 \cdot \cos(-x)$

$= -x^3 \cdot \cos(x) = -f(x)$

PROBLEM Some Functions that are neither even nor odd

1. $f(x) = x^3 + x^2$

2. $f(x) = 1 + \sin(x)$

1. $f(-x)$

$= (-x)^3 + (-x)^2$

$= -x^3 + x^2$

$f(x) = x^3 + x^2$

$-f(x) = -x^3 - x^2$

2.

$f(-x) = 1 + \sin(-x)$

$= 1 - \sin x$

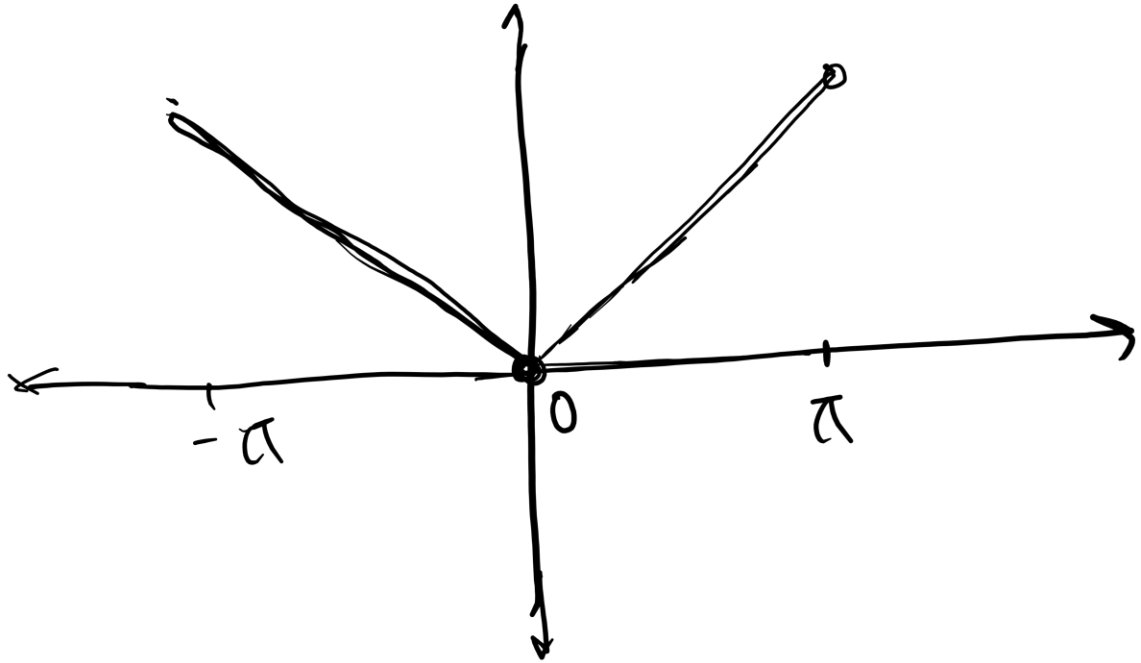
$f(x) = 1 + \sin x$

$-f(x) = -1 - \sin x$

PROBLEM Determine whether the functions are odd, even or neither.

V.V.G

$$f(x) = \begin{cases} -x & -\pi < x < 0 \\ x & 0 < x < \pi \end{cases}$$

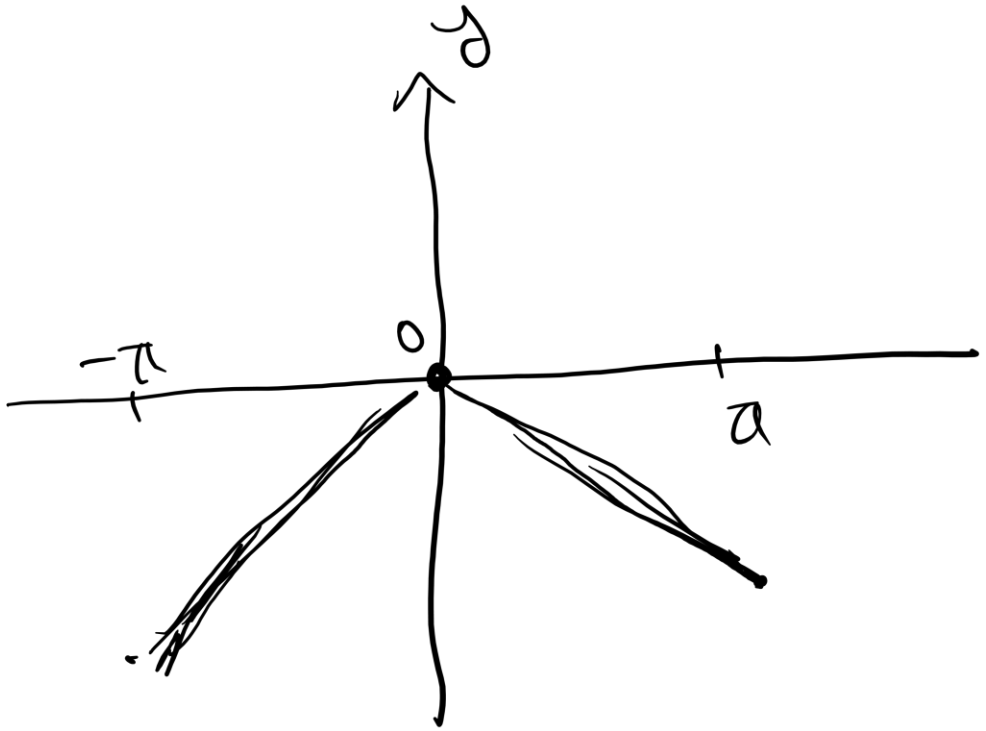


Even

PROBLEM Determine whether the functions are odd, even or neither.

v.v.g

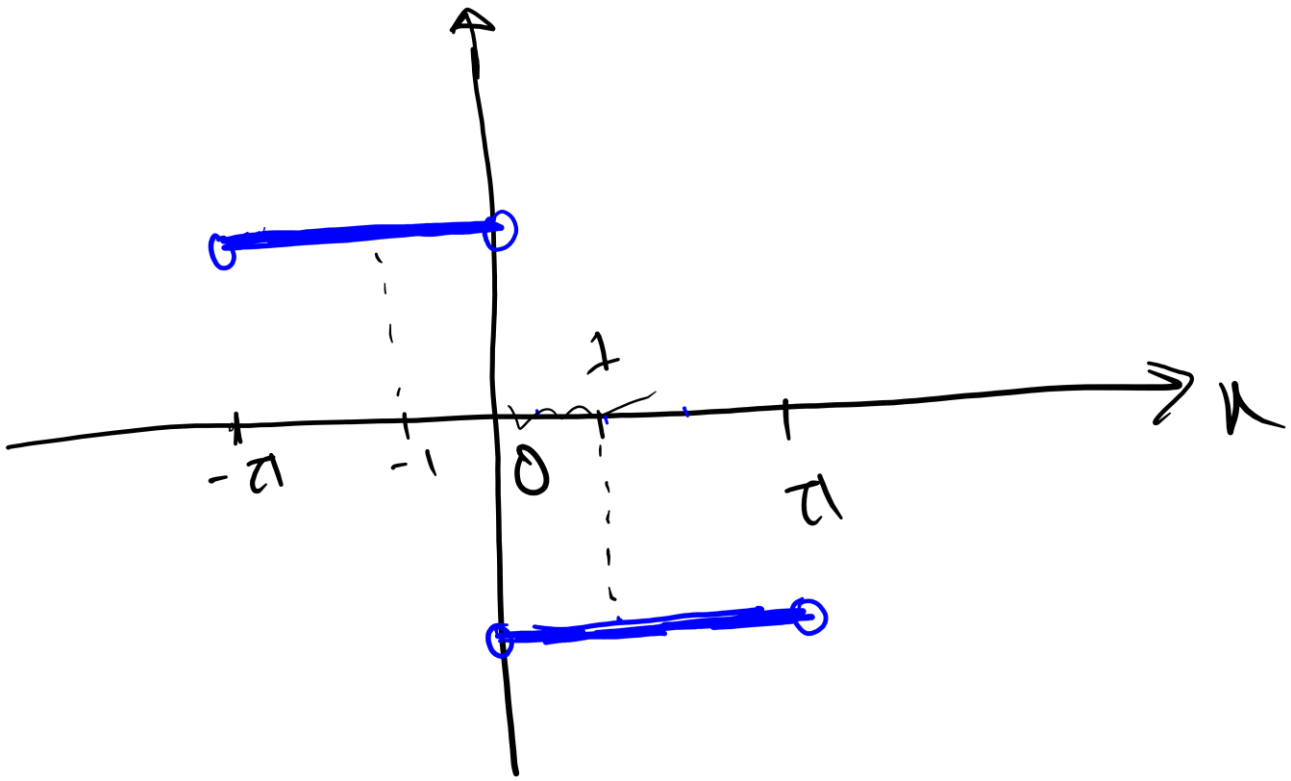
$$f(x) = \begin{cases} x & -\pi < x < 0 \\ -x & 0 < x < \pi \end{cases}$$



Even

PROBLEM Determine whether the functions are odd, even or neither.

$$f(x) = \begin{cases} 2 & -\pi < x < 0 \\ -2 & 0 < x < \pi \end{cases}$$

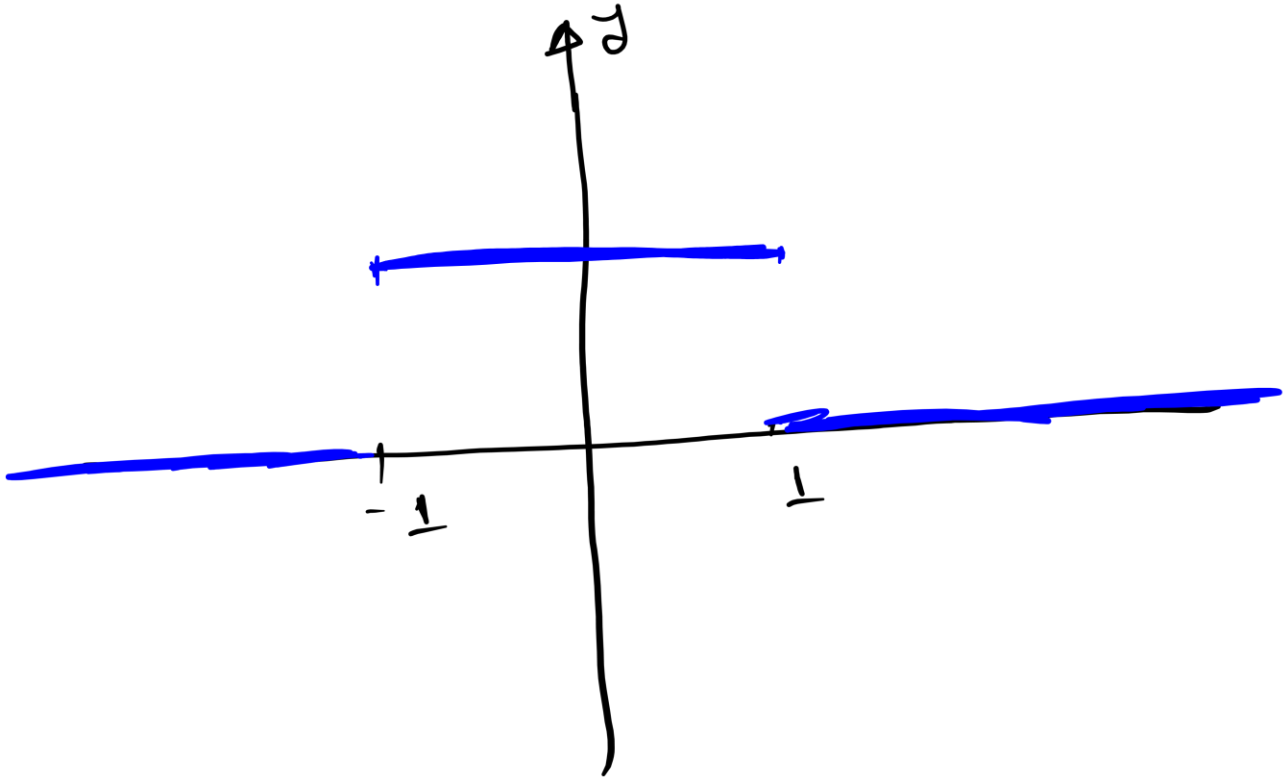


Odd

PROBLEM Determine whether the functions are odd, even or neither.

$$f(x) = \begin{cases} 1 & |x| < 1 \\ 0 & otherwise \end{cases}$$

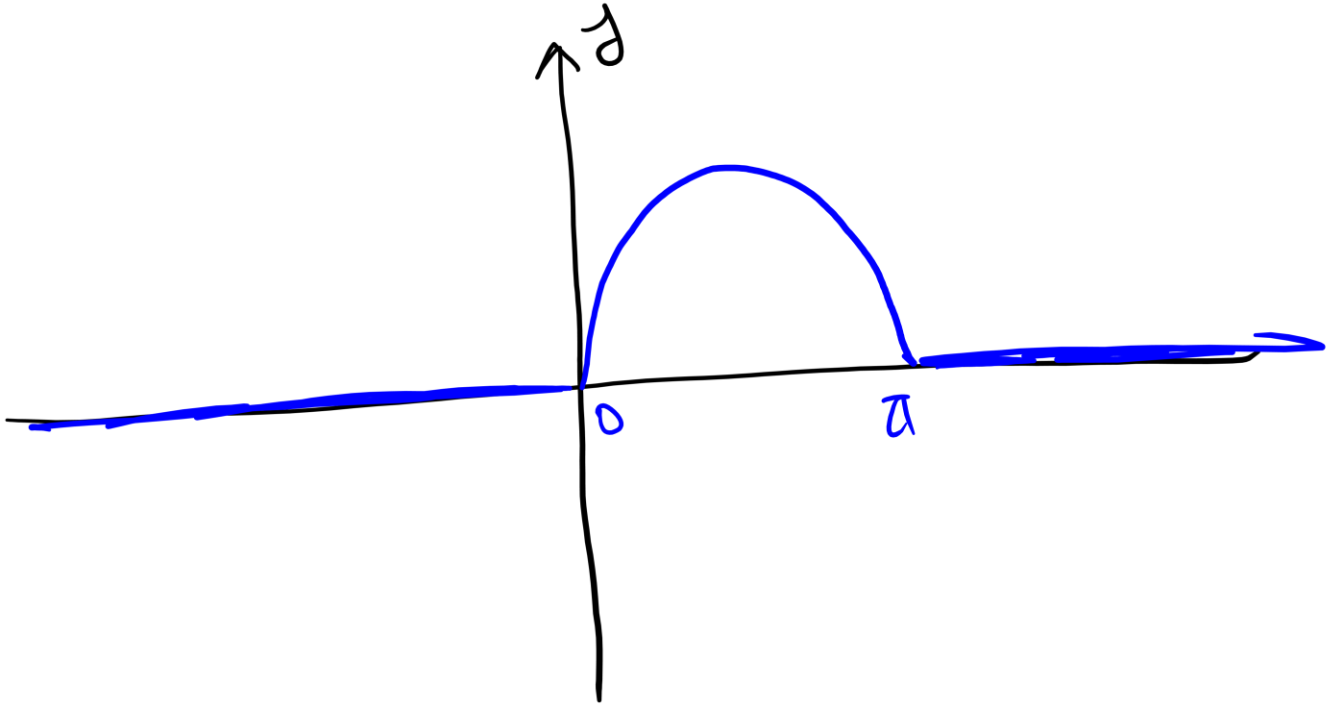
$-1 < x < 1$



Even.

PROBLEM Determine whether the functions are odd, even or neither.

$$f(x) = \begin{cases} 2\sin x & 0 < x < \pi \\ 0 & otherwise \end{cases}$$



neither Even

nor odd

Trigonometric Identity

Trigonometric Identities

$$\sin(A+B) = \sin A \cos B + \cos A \sin B$$

$$\sin(A-B) = \sin A \cos B - \cos A \sin B$$

$$\cos(A+B) = \cos A \cos B - \sin A \sin B$$

$$\cos(A-B) = \cos A \cos B + \sin A \sin B$$

Trigonometric Identities (V.V.g)

$$2\sin A\sin B = \cos(A-B) - \cos(A+B)$$

$$2\cos A\cos B = \cos(A+B) + \cos(A-B)$$

$$2\sin A\cos B = \sin(A+B) + \sin(A-B)$$

$$2\sin^2 A = 1 - \cos 2A$$

$$2\cos^2 A = 1 + \cos 2A$$

Trigonometric Identities (V.V.9)

$\sin(k \cdot \pi) = 0$ for all integer k

$\cos(k \cdot \pi) = (-1)^k$ for all integer k

$\sin(\text{integer} \cdot \pi) = 0$

$\cos(\text{integer} \cdot \pi) = (-1)^{\text{integer}}$

Integration

The uv Formula of Integration (Integration by Parts)

$$\int u \cdot v \cdot dx = u \int v\, dx - \int \left(\frac{du}{dx} \cdot \int v\, dx \right) dx.$$

$$\int \underline{x} \cdot \underline{e^{2x}}\, dx = x \cdot \int e^{2x} - \int \left(\frac{d}{dx}(x) \cdot \int e^{2x} dx \right) dx$$

$$= x \cdot \frac{e^{2x}}{2} - \int 1 \cdot \frac{e^{2x}}{2}\, dx = \frac{1}{2} x e^{2x} - \frac{1}{4} e^{2x} + C.$$

The uv Formula of Integration (Integration by Parts Shortcut)

$$\int (x^3+2x)\cos(3x)\,dx$$

	u	v
$+$	x^3+2x	$\cos(3x)$
$-$	$3x^2+2$	$\frac{1}{3}\sin 3x$
$+$	$6x$	$-\frac{1}{9}\cos 3x$
$-$	6	$-\frac{1}{27}\sin 3x$
$+$	0	$\frac{1}{81}\cos 3x$

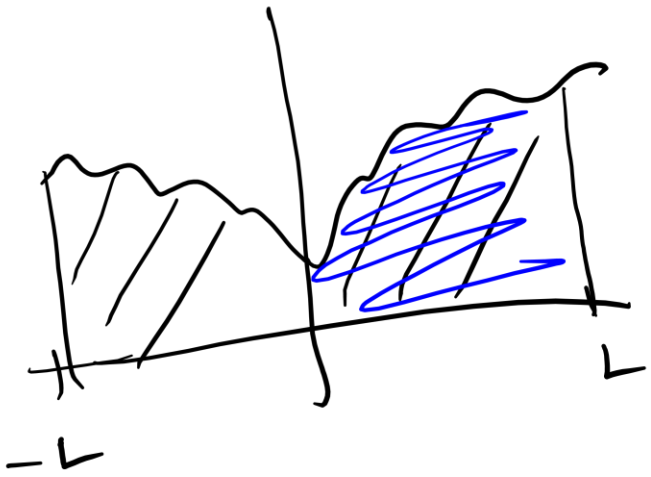
$$= (x^3+2x)\frac{1}{3}\sin 3x - (3x^2+2)\frac{-1}{9}\cos 3x$$

$$+ (6x)\frac{-1}{27}\sin 3x - 6\cdot\frac{1}{81}\cos 3x$$

Integration of Even Functions on $[-L, L]$

$$\int_{-L}^{L} f(x)\,dx = 2 \cdot \int_{0}^{L} f(x)\,dx$$

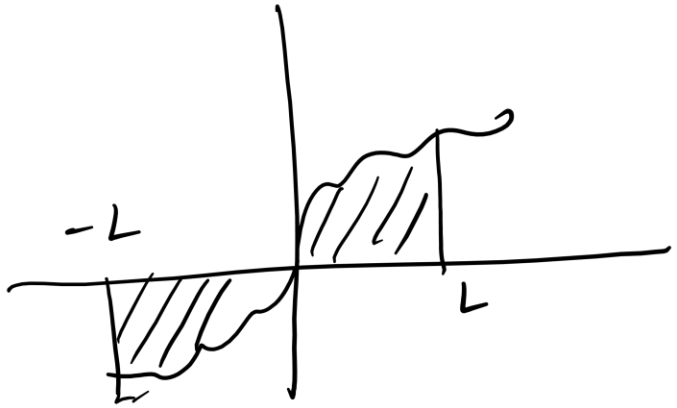
Even



Integration of Odd Functions on $[-L, L]$

$$\int_{-L}^{L} f(x)\,dx = 0.$$

odd



Thank you!